

SERVICE MANUAL

Invivo Corporation
3545 SW 47th Avenue
Gainesville, FL 32608

Document TR0528S

Revision 03

GE 1.5T 32-Channel Cardiac Coil

GE Part Number: M7000LB

Invivo Part Number: 800528

Manufactured By:



© Invivo Corporation, 2009

Distributed By:



GE Healthcare

Damage in Transportation

All packages should be closely examined at time of delivery. If damage is apparent, have notation “**damage in shipment**” written on **all** copies of the freight or express bill **before** delivery is accepted or “signed for” by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

File a report with

- Call 1–800–548–3366 and use option 8.
- Fill out a report on <http://egems.med.ge.com/edq/home.jsp>
- Contact your local service coordinator for more information on this process.

Rev. January 7, 2010

LANGUAGE POLICY FOR SERVICE DOCUMENTATION (DOC0620154 REV 1)

ПРЕДУПРЕЖДЕНИЕ (BG)	Това упътване за работа е налично само на английски език. - Ако доставчикът на услугата на клиента изиска друг език, задължение на клиента е да осигури превод. - Не използвайте оборудването, преди да сте се консултирали и разбрали упътването за работа. - Неспазването на това предупреждение може да доведе до нараняване на доставчика на услугата, оператора или пациента в резултат на токов удар, механична или друга опасност.
警告 (ZH-CN)	本维修手册仅提供英文版本。 - 如果客户的维修服务人员需要非英文版本，则客户需自行提供翻译服务。 - 未详细阅读和完全理解本维修手册之前，不得进行维修。 - 忽略本警告可能对维修服务人员、操作人员或患者造成电击、机械伤害或其他形式的伤害。
警告 (ZH-HK)	本服務手冊僅提供英文版本。 - 倘若客戶的服務供應商需要英文以外之服務手冊，客戶有責任提供翻譯服務。 - 除非已參閱本服務手冊及明白其內容，否則切勿嘗試維修設備。 - 不遵從本警告或會令服務供應商、網絡供應商或病人受到觸電、機械性或其他危險。
警告 (ZH-TW)	本維修手冊僅有英文版。 - 若客戶的維修廠商需要英文版以外的語言，應由客戶自行提供翻譯服務。 - 請勿試圖維修本設備，除非 您已查閱並瞭解本維修手冊。 - 若未留意本警告，可能導致維修廠商、操作員或病患因觸電、機械或其他危險而受傷。
UPOZORENJE (HR)	Ovaj servisni priručnik dostupan je na engleskom jeziku. - Ako davatelj usluge klijenta treba neki drugi jezik, klijent je dužan osigurati prijevod. - Ne pokušavajte servisirati opremu ako niste u potpunosti pročitali i razumjeli ovaj servisni priručnik. - Zanemarite li ovo upozorenje, može doći do ozljede davatelja usluge, operatera ili pacijenta uslijed strujnog udara, mehaničkih ili drugih rizika.
VÝSTRAHA (CS)	Tento provozní návod existuje pouze v anglickém jazyce. - V případě, že externí služba zákazníkům potřebuje návod v jiném jazyce, je zajištění překladu do odpovídajícího jazyka úkolem zákazníka. - Nesnažte se o údržbu tohoto zařízení, aniž byste si přečetli tento provozní návod a pochopili jeho obsah. - V případě nedodržování této výstrahy může dojít k poranění pracovníka prodejního servisu, obslužného personálu nebo pacientů vlivem elektrického proudu, respektive vlivem mechanických či jiných rizik.
ADVARSEL (DA)	Denne servicemanual findes kun på engelsk. - Hvis en kundes tekniker har brug for et andet sprog end engelsk, er det kundens ansvar at sørge for oversættelse. - Forsøg ikke at servicere udstyret uden at læse og forstå denne servicemanual. - Manglende overholdelse af denne advarsel kan medføre skade på grund af elektrisk stød, mekanisk eller anden fare for tekniker, operatøren eller patienten.
WAARSCHUWING (NL)	Deze onderhoudshandleiding is enkel in het Engels verkrijgbaar. - Als het onderhoudspersoneel een andere taal vereist, dan is de klant verantwoordelijk voor de vertaling ervan. - Probeer de apparatuur niet te onderhouden alvorens deze onderhoudshandleiding werd geraadpleegd en begrepen is. - Indien deze waarschuwing niet wordt opgevolgd, zou het onderhoudspersoneel, de operator of een patiënt gewond kunnen raken als gevolg van een elektrische schok, mechanische of andere gevaren.

WARNING (EN)	This service manual is available in English only. • If a customer's service provider requires a language other than english, it is the customer's responsibility to provide translation services. • Do not attempt to service the equipment unless this service manual has been consulted and is understood. • Failure to heed this warning may result in injury to the service provider, operator or patient from electric shock, mechanical or other hazards.
HOIATUS (ET)	See teenindusjuhend on saadaval ainult inglise keeles • Kui klienditeeninduse osutaja nõuab juhendit inglise keelest erinevas keeles, vastutab klient tõlketeenuse osutamise eest. • Ärge üritage seadmeid teenindada enne eelnevalt käesoleva teenindusjuhendiga tutvumist ja sellest aru saamist. • Käesoleva hoiatuse eiramine võib põhjustada teenuseosutaja, operaatori või patsiendi vigastamist elektrilöögi, mehaanilise või muu ohu tagajärjel.
VAROITUS (FI)	Tämä huolto-ohje on saatavilla vain englanniksi. • Jos asiakkaan huoltohenkilöstö vaatii muuta kuin englanninkielistä materiaalia, tarvittavan käännöksen hankkiminen on asiakkaan vastuulla. • Älä yritä korjata laitteistoa ennen kuin olet varmasti lukenut ja ymmärtänyt tämän huolto-ohjeen. • Mikäli tätä varoitusta ei noudateta, seurauksena voi olla huoltohenkilöstön, laitteiston käyttäjän tai potilaan vahingoittuminen sähköiskun, mekaanisen vian tai muun vaaratilanteen vuoksi.
ATTENTION (FR)	Ce manuel d'installation et de maintenance est disponible uniquement en anglais. • Si le technicien d'un client a besoin de ce manuel dans une langue autre que l'anglais, il incombe au client de le faire traduire. • Ne pas tenter d'intervenir sur les équipements tant que ce manuel d'installation et de maintenance n'a pas été consulté et compris. • Le non-respect de cet avertissement peut entraîner chez le technicien, l'opérateur ou le patient des blessures dues à des dangers électriques, mécaniques ou autres.
WARNUNG (DE)	Diese Serviceanleitung existiert nur in englischer Sprache. • Falls ein fremder Kundendienst eine andere Sprache benötigt, ist es Aufgabe des Kunden für eine entsprechende Übersetzung zu sorgen. • Versuchen Sie nicht diese Anlage zu warten, ohne diese Serviceanleitung gelesen und verstanden zu haben. • Wird diese Warnung nicht beachtet, so kann es zu Verletzungen des Kundendiensttechnikers, des Bedieners oder des Patienten durch Stromschläge, mechanische oder sonstige Gefahren kommen.
ΠΡΟΕΙΔΟΠΟΙΗΣΗ (EL)	Το παρόν εγχειρίδιο σέρβις διατίθεται μόνο στα αγγλικά. • Εάν ο τεχνικός σέρβις ενός πελάτη απαιτεί το παρόν εγχειρίδιο σε γλώσσα εκτός των αγγλικών, αποτελεί ευθύνη του πελάτη να παρέχει τις υπηρεσίες μετάφρασης. • Μην επιχειρήσετε την εκτέλεση εργασιών σέρβις στον εξοπλισμό αν δεν έχετε συμβουλευτεί και κατανοήσει το παρόν εγχειρίδιο σέρβις. • Αν δεν προσέξετε την προειδοποίηση αυτή, ενδέχεται να προκληθεί τραυματισμός στον τεχνικό σέρβις, στο χειριστή ή στον ασθενή από ηλεκτροπληξία, μηχανικούς ή άλλους κινδύνους.
FIGYELMEZTETÉS (HU)	Ezen karbantartási kézikönyv kizárólag angol nyelven érhető el. • Ha a vevő szolgáltatója angoltól eltérő nyelvre tart igényt, akkor a vevő felelőssége a fordítás elkészítése. • Ne próbálja elkezdni használni a berendezést, amíg a karbantartási kézikönyvben leírtakat nem értelmezték. • Ezen figyelmeztetés figyelmen kívül hagyása a szolgáltató, működtető vagy a beteg áramütés, mechanikai vagy egyéb veszélyhelyzet miatti sérülését eredményezheti.
ÆDVÖRUN (IS)	Þessi þjónustuhandbók er aðeins fáanleg á ensku. • Ef að þjónustuveitandi viðskiptamanns þarfnast annas tungumáls en ensku, er það skylda viðskiptamanns að skaffa tungumálaþjónustu. • Reynið ekki að afgreiða tækið nema að þessi þjónustuhandbók hefur verið skoðuð og skilin. • Brot á sinna þessari æðvörun getur leitt til meiðsla á þjónustuveitanda, stjórnanda eða sjúklings frá raflosti, vélrænu eða öðrum áhættum.
AVVERTENZA (IT)	Il presente manuale di manutenzione è disponibile soltanto in lingua inglese. • Se un addetto alla manutenzione richiede il manuale in una lingua diversa, il cliente è tenuto a provvedere direttamente alla traduzione. • Procedere alla manutenzione dell'apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto. • Il mancato rispetto della presente avvertenza potrebbe causare lesioni all'addetto alla manutenzione, all'operatore o ai pazienti provocate da scosse elettriche, urti meccanici o altri rischi.

警告 (JA)	このサービスマニュアルには英語版しかありません。 • サービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。 • このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないでください。 • この警告に従わない場合、サービスを担当される方、操作員あるいは患者 さんが、感電や機械的又はその他の危険により負傷する可能性があります。
경고 (KO)	본 서비스 매뉴얼은 영어로만 이용하실 수 있습니다. • 고객의 서비스 제공자가 영어 이외의 언어를 요구할 경우, 번역 서비스를 제공하는 것은 고객의 책임입니다. • 본 서비스 매뉴얼을 참조하여 숙지하지 않은 이상 해당 장비를 수리하려고 시도하지 마십시오. • 본 경고 사항에 유의하지 않으면 전기 쇼크, 기계적 위험, 또는 기타 위험으로 인해 서비스 제공자, 사용자 또는 환자에게 부상을 입힐 수 있습니다.
BRĪDINĀJUMS (LV)	Šī apkopes rokasgrāmata ir pieejama tikai angļu valodā. • Ja klienta apkopes sniedzējam nepieciešama informācija citā valodā, klienta pienākums ir nodrošināt tulkojumu. • Neveiciet aprikojuma apkopi bez apkopes rokasgrāmatas izlasīšanas un saprašanas. • Šī brīdinājuma neievērošanas rezultātā var rasties elektriskās strāvas trieciena, mehānisku vai citu faktoru izraisītu traumu risks apkopes sniedzējam, operatoram vai pacientam.
ĮSPĖJIMAS (LT)	Šis eksploatavimo vadovas yra tik anglų kalba. • Jei kliento paslaugų tiekėjas reikalauja vadovo kita kalba – ne anglų, suteikti vertimo paslaugas privalo klientas. • Nemėginkite atlikti įrangos techninės priežiūros, jei neperskaitėte ar nesupratote šio eksploatavimo vadovo. • Jei nepaisysite šio įspėjimo, galimi paslaugų tiekėjo, operatoriaus ar paciento sužalojimai dėl elektros šoko, mechaninių ar kitų pavojų.
ADVARSEL (NO)	Denne servicehåndboken finnes bare på engelsk. • Hvis kundens serviceleverandør har bruk for et annet språk, er det kundens ansvar å sørge for oversettelse. • Ikke forsøk å reparere utstyret uten at denne servicehåndboken er lest og forstått. • Manglende hensyn til denne advarselen kan føre til at serviceleverandøren, operatøren eller pasienten skades på grunn av elektrisk støt, mekaniske eller andre farer.
OSTRZEŻENIE (PL)	Niniejszy podręcznik serwisowy dostępny jest jedynie w języku angielskim. • Jeśli serwisant klienta wymaga języka innego niż angielski, zapewnienie usługi tłumaczenia jest obowiązkiem klienta. • Nie próbować serwisować urządzenia bez zapoznania się z niniejszym podręcznikiem serwisowym i zrozumienia go. • Niezastosowanie się do tego ostrzeżenia może doprowadzić do obrażeń serwisanta, operatora lub pacjenta w wyniku porażenia prądem elektrycznym, zagrożenia mechanicznego bądź innego.

AVISO (PT-BR)	Este manual de assistência técnica encontra-se disponível unicamente em inglês. • Se outro serviço de assistência técnica solicitar a tradução deste manual, caberá ao cliente fornecer os serviços de tradução. • Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica. • A não observância deste aviso pode ocasionar ferimentos no técnico, operador ou paciente decorrentes de choques elétricos, mecânicos ou outros.
ATENÇÃO (PT-PT)	Este manual de assistência técnica só se encontra disponível em inglês. • Se qualquer outro serviço de assistência técnica solicitar este manual noutro idioma, é da responsabilidade do cliente fornecer os serviços de tradução. • Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica. • O não cumprimento deste aviso pode colocar em perigo a segurança do técnico, do operador ou do paciente devido a choques eléctricos, mecânicos ou outros.
ATENȚIE (RO)	Acest manual de service este disponibil doar în limba engleză. • Dacă un furnizor de servicii pentru clienți necesită o altă limbă decât cea engleză, este de datoria clientului să furnizeze o traducere. • Nu încercați să reparați echipamentul decât ulterior consultării și înțelegerii acestui manual de service. • Ignorarea acestui avertisment ar putea duce la rănirea depanatorului, operatorului sau pacientului în urma pericolelor de electrocutare, mecanice sau de altă natură.
ОСТОРОЖНО! (RU)	Данное руководство по техническому обслуживанию представлено только на английском языке. • Если сервисному персоналу клиента необходимо руководство не на английском, а на каком-то другом языке, клиенту следует самостоятельно обеспечить перевод. • Перед техническим обслуживанием оборудования обязательно обратитесь к данному руководству и поймите изложенные в нем сведения. • Несоблюдение требований данного предупреждения может привести к тому, что специалист по техобслуживанию, оператор или пациент получит удар электрическим током, механическую травму или другое повреждение.
UPOZORENJE (SR)	Ovo servisno uputstvo je dostupno samo na engleskom jeziku. • Ako klijentov serviser zahteva neki drugi jezik, klijent je dužan da obezbedi prevodilačke usluge. • Ne pokušavajte da opravite uređaj ako niste pročitali i razumeli ovo servisno uputstvo. • Zanemarivanje ovog upozorenja može dovesti do povređivanja serviser, rukovaoca ili pacijenta usled strujnog udara ili mehaničkih i drugih opasnosti.
UPOZORNENIE (SK)	Tento návod na obsluhu je k dispozícii len v angličtine. • Ak zákazníkovi poskytovateľ služieb vyžaduje iný jazyk ako angličtinu, poskytnutie prekladateľských služieb je zodpovednosťou zákazníka. • Nepokúšajte sa o obsluhu zariadenia, kým si neprečítate návod na obsluhu a neporozumiete mu. • Zanedbanie tohto upozornenia môže spôsobiť zranenie poskytovateľa služieb, obsluhujúcej osoby alebo pacienta elektrickým prúdom, mechanické alebo iné ohrozenie.
ATENCION (ES)	Este manual de servicio sólo existe en inglés. • Si el encargado de mantenimiento de un cliente necesita un idioma que no sea el inglés, el cliente deberá encargarse de la traducción del manual. • No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio. • La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.
VARNING (SV)	Den här servicehandboken finns bara tillgänglig på engelska. . • Om en kunds servicetekniker har behov av ett annat språk än engelska, ansvarar kunden för att tillhandahålla översättningstjänster. • Försök inte utföra service på utrustningen om du inte har läst och förstår den här servicehandboken. • Om du inte tar hänsyn till den här varningen kan det resultera i skador på serviceteknikern, operatören eller patienten till följd av elektriska stötar, mekaniska faror eller andra faror.
DİKKAT (TR)	Bu servis kılavuzunun sadece ingilizcesi mevcuttur. • Eğer müşteri teknisyeni bu kılavuzu ingilizce dışında bir başka lisandan talep ederse, bunu tercüme ettirmek müşteriye düşer. • Servis kılavuzunu okuyup anlamadan ekipmanlara müdahale etmeyiniz. • Bu uyarıya uyulmaması, elektrik, mekanik veya diğer tehlikelerden dolayı teknisyen, operatör veya hastanın yaralanmasına yol açabilir.

TABLE OF CONTENTS

SECTION 1 – INTRODUCTION.....	9
1.1 Product Identification and Shipping List	9
1.2 Shipping Contents.....	10
1.3 Compatibility	10
1.4 Related Documentation.....	10
1.5 Environmental Requirements	10
SECTION 2 – INSTALLATION PROCEDURE	11
2.1 Personnel Requirements.....	11
2.2 Tools and Test Equipment.....	11
2.3 Safety – Not applicable.....	11
2.4 Required Conditions.....	11
2.5 Procedure.....	11
2.6 Finalization	15
SECTION 3: 32CH CARDIAC COIL MCQA TEST	16
3.1 Tools Required	16
3.2 MCQA Setup	16
3.3 MCQA Test	18
3.4 Periodic Quality Assurance Check	19
SECTION 4: TROUBLESHOOTING	19
4-1 Mechanical Hardware Check	19
4-2 Electrical Hardware check.....	19
4-3 Coil ID Troubleshooting Tips	21
SECTION 5 – MAINTENANCE	22
5-1 Coil Care.....	22
SECTION 6 – RENEWAL PARTS	22
6.1 Field Replaceable Units	22
6.2 Other Replaceable Accessories	23

ATTENTION!

A GE field engineer **MUST** perform installation of this coil. Please contact your field engineer to arrange for installation.

Attempts to perform installation of this coil without proper system configuration will result in system errors.

SECTION 1 – INTRODUCTION

1.1 Product Identification and Shipping List

This is a service manual for the GE 1.5T 32-Channel Cardiac Coil. The 32-channel Cardiac Coil consists of two 16-channel coil halves (anterior & posterior). Each half is uniquely identified by using the labels below:

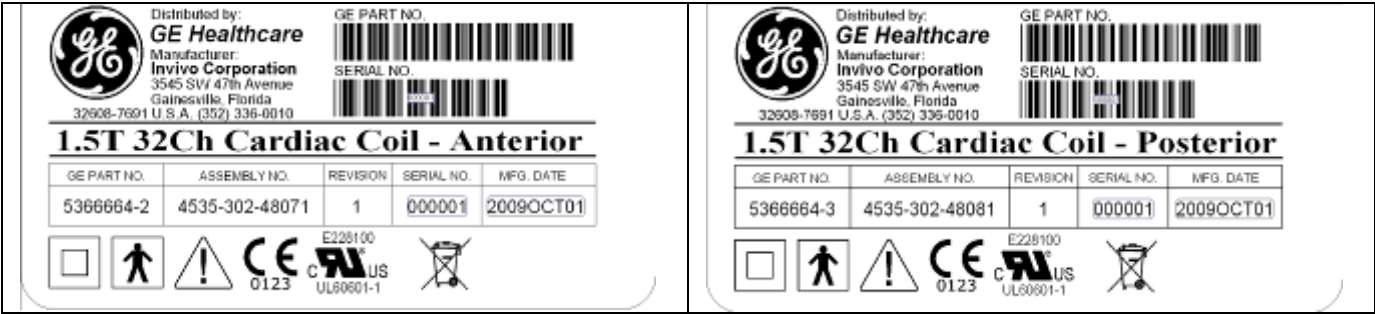


Illustration 1: Cardiac Coil Labels

Illustration 2 shows images of each half of the 1.5T 32 Channel Cardiac Coil

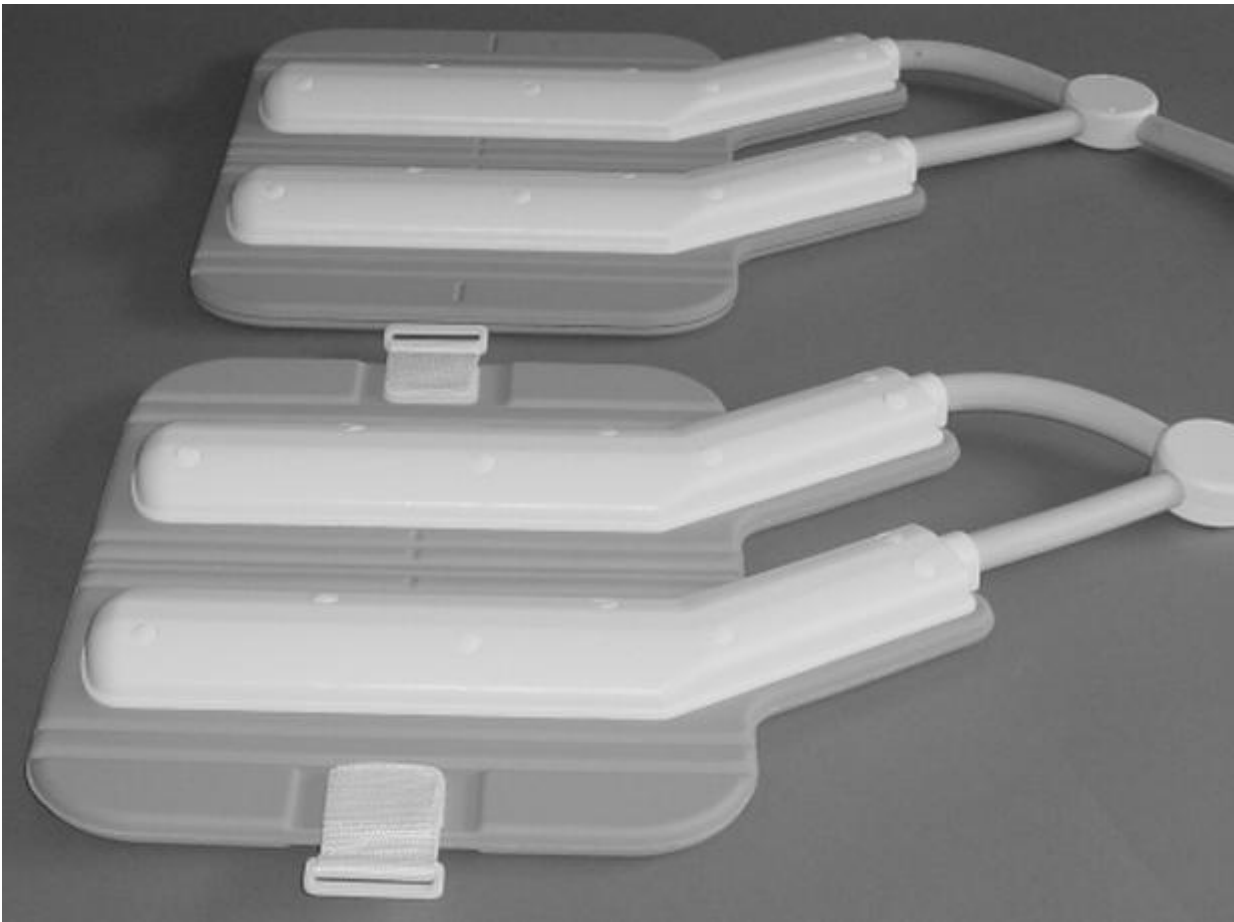


Illustration 2

1.2 Shipping Contents

SHIPPING LIST BOX 1– TABLE 1-2-1

Description	Invivo Part #	GE Part #	Qty
1.5T 32Ch Cardiac Coil - Anterior	4535-302-48071	5366664-2	1
1.5T 32Ch Cardiac Coil - Posterior	4535-302-48081	5366664-3	1
32Ch Cardiac Coil - Phantom Positioner (2 pieces)	4535-302-66001	5366627-4	1
32Ch Cardiac Coil - Coil Definition Disk	N/A	5366627-5	1
32Ch Cardiac Coil - Posterior Pad, Bottom	4535-302-58891	N/A	1
32Ch Cardiac Coil - Posterior Pad, Top	4535-302-58901		1
32Ch Cardiac Coil - Head Pad	4535-302-58911	N/A	1
32Ch Cardiac Coil - Foot Pad	4535-302-58921	N/A	1
32Ch Cardiac Coil - Anterior Pad	4535-302-60711	N/A	1
32Ch Cardiac Coil - Operator Manual Insert	4535-302-85131	N/A	1
32Ch Cardiac Coil - Service Manual	TR0528S	N/A	1

1.3 Compatibility

This coil is compatible with the following hardware configurations:

Discovery MR450 and Optima MR450w 1.5T Systems

1.4 Related Documentation

- 5500110 Discovery MR450 1.5T Service Methods DVD
- 5670000 Optima MR450w 1.5T Service Methods DVD
- 1.5T & 3.0T 32-Channel Cardiac Coil Operator's Manual, 4535-302-58991

1.5 Environmental Requirements

Storage

Store the coil in an air-conditioned scan room or equipment room. To store the coils, a storage space of 48.3 cm/19.0 in (width) x 16.5 cm/6.5 in (depth) x 101.6 cm/40.0 (length) is required.

Dimensions

Coil (with posterior pad):	48.3 cm x 16.5 cm x 101.6 cm	(19.0 in x 6.5 in x 40.0 in)
Phantom Positioner:	50.6 cm x 50.6 cm x 17.7 cm	(20 in x 20 in x 7 in)

Weight

Coil half:	2.95 Kg	(6.5 lb)
Phantom Positioner:	0.68 Kg	(1.5 lb)

SECTION 2 – INSTALLATION PROCEDURE

2.1 Personnel Requirements

Required Persons	Procedure	Finalization
1	15 mins	45 minutes

2.2 Tools and Test Equipment

Item	Qty	Part#
32Ch Cardiac Coil – Coil Definition Disk	1	5366627-5
Phantoms and phantom positioners are needed for the MCQA test.	N/a	See section 3

2.3 Safety – Not applicable

2.4 Required Conditions

Condition
For 450: The minimum software build to install this coil is 20.1 M3 or 20.1_IB1
For 450w: The minimum software build to install this coil is 21.0 M3

2.5 Procedure

NOTE: Do not plug in the coil and use AutoCoil to install the coil. This method will not install the coil protocols.

- 2.5.1 Make sure that the latest software build is loaded on the system prior to installing the Cardiac coil. The 32Ch Cardiac protocols will only install if the required conditions are met.
- 2.5.2 Check to confirm if the 32chCardiac coil is already installed on the system software by opening the Service Browser under the Service Desktop Manager. In the Service Browser click on the Configuration tab, then the Coil Database Explorer under the configuration folder on the left.
- 2.5.3 Select “Click here to start this tool”. The Coil DB Explorer window will open and under the "Coil Name" tab there is a "Clinical Coil Name" control. Click on this and there will be a dropdown list of all the logical coil names. There should be three names that start with "C-GE_32Ch Cardiac". This indicates the coil is present in the Coil DB. Reference Illustration 3. If the coil is not present on the list proceed to step 2.5.6.

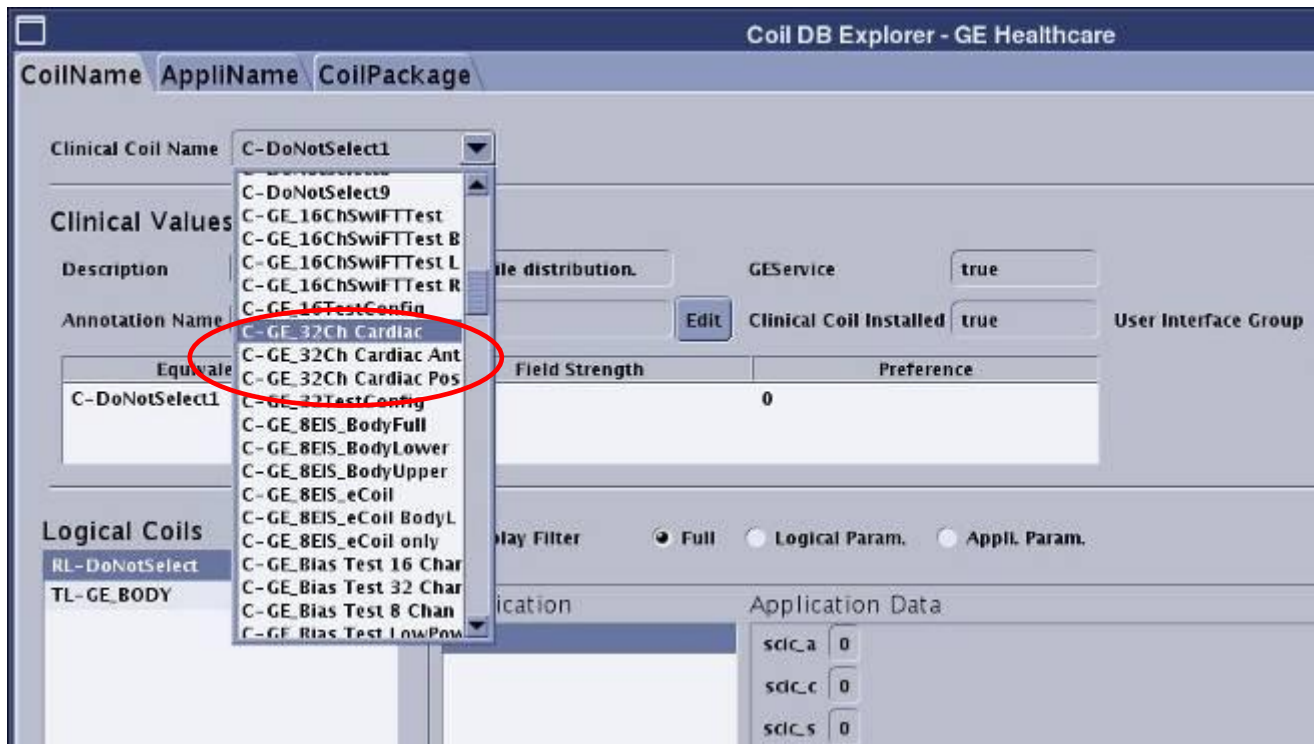


Illustration 3: Confirmation of coil configurations installed

- 2.5.4 The 32Ch Cardiac coil also has two coil protocols that must be on the system. Open the Protocol Management tab. Confirm the coil protocols are installed by viewing the “Chest” protocols under the GE Protocol Library. Two protocols that end in “- 32ch Cardiac” must be present as shown in Illustration 4.

If protocols are not present and the coil CD was just installed one may have to exit the “Chest” protocols by selecting another area such as “Abdomen” and then reselecting “Chest” for the two protocols to appear.

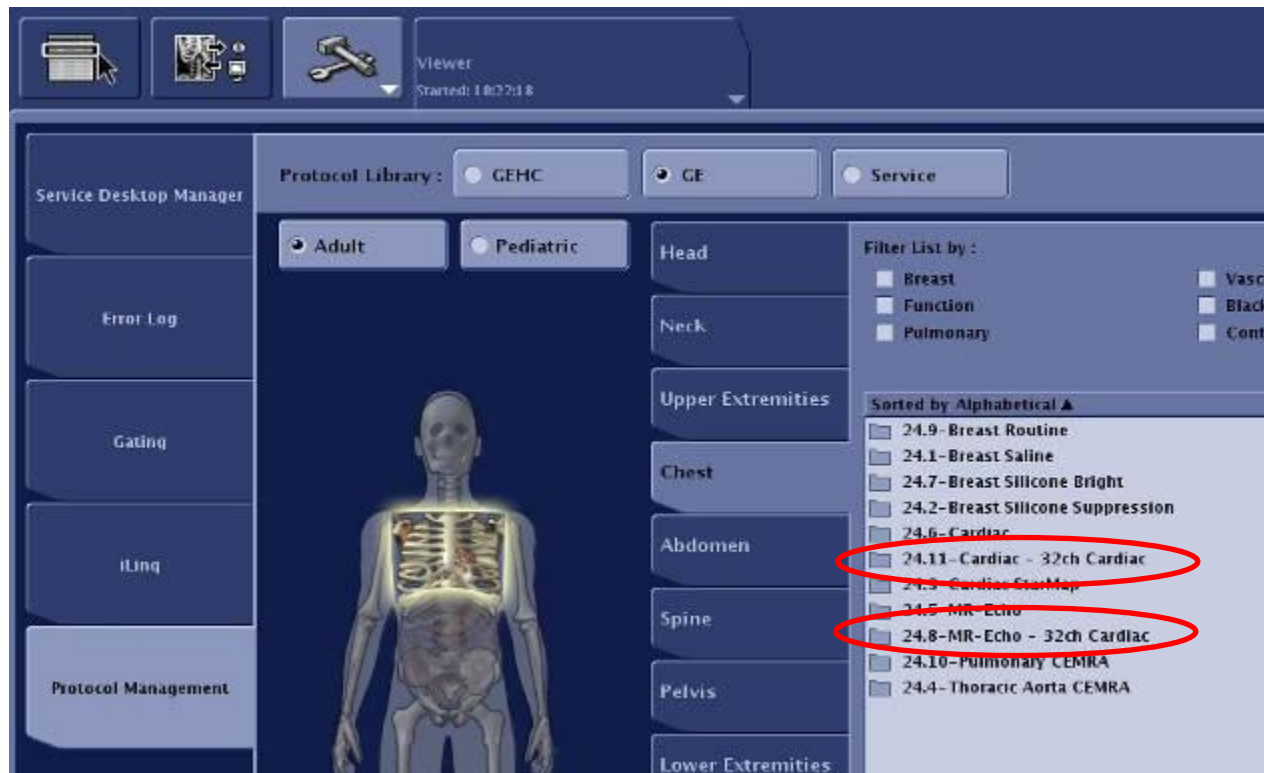


Illustration 4: Confirmation of 32ch Cardiac coil protocols installed

- 2.5.5 If the cardiac coil configurations and protocols are on the system skip to the finalization steps in section 2-6.
- 2.5.6 To install the cardiac coil configurations and protocols, insert the Cardiac Coil Definitions Disc into the default (typically upper) host DVD drive.
- 2.5.7 Open a C-Shell from the Service Desktop manager and type the following commands:

su

```
<root password>  
mount /mnt/cdrom  
/mnt/cdrom/autorun
```

- 2.5.8 A Coil Data Disk Install window will open. Type “i” to install the CD contents.
- 2.5.9 When the contents of the CD installation is complete, the cursor will appear and you will see the message as shown in Illustration 5. Press enter to exit the Coil CD installation.

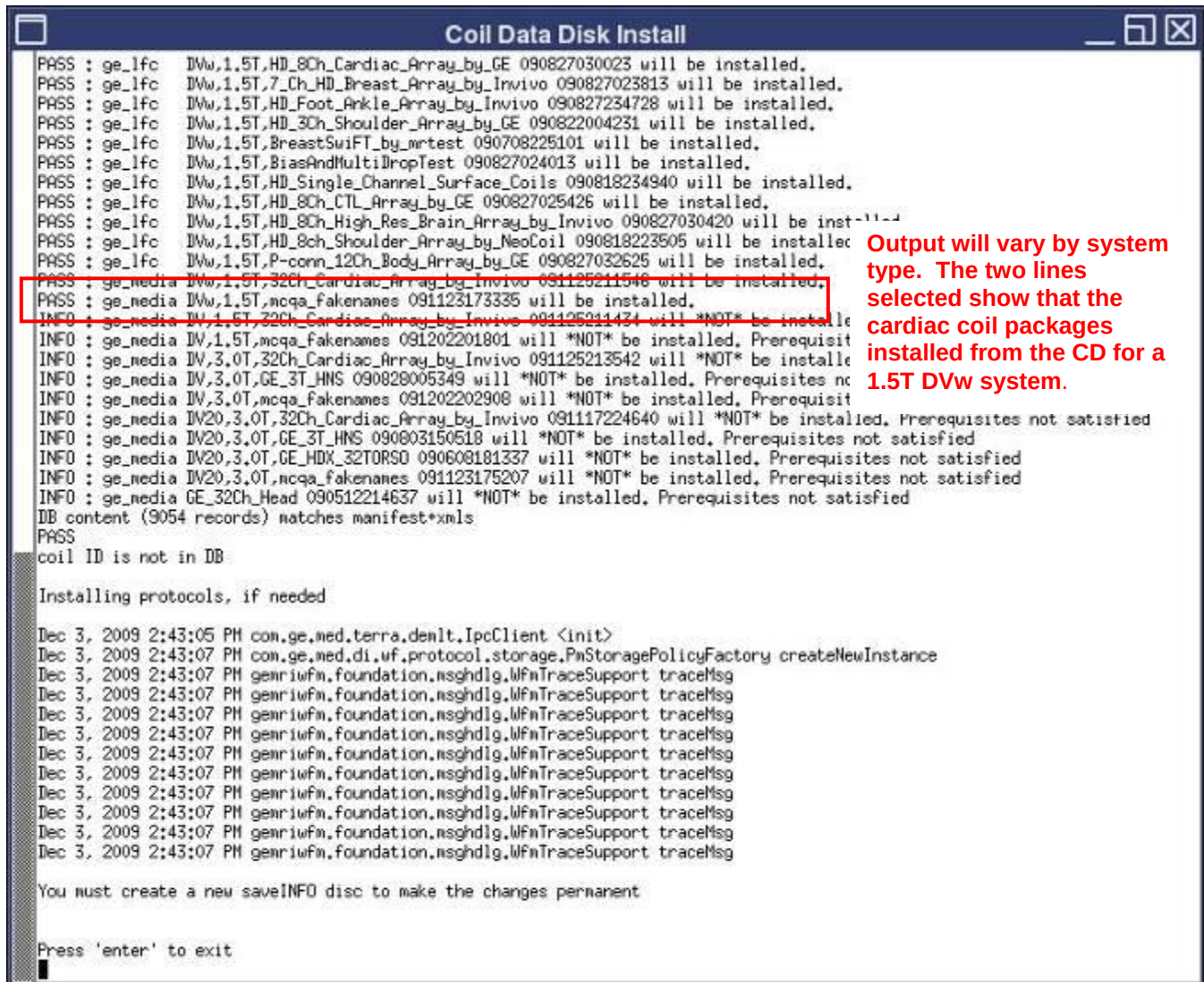


Illustration 5: CD Contents Installation Complete

- 2.5.10 Type <eject> in the original c shell. Remove the CD from the host DVD drive.
- 2.5.11 To confirm the coil has been properly installed from the coil CD follow steps 2.5.2 – 2.5.4.

2.6 Finalization

- 2.6.1 Perform a SaveInfo under Guided Install to make the changes permanent.
- 2.6.2 Perform the Multi-Coil Quality Assurance (MCQA) test with this coil. Refer to Section 3.
- 2.6.3 Complete Product Locator information for serialized components. Failure to do this may result in failure of your site to receive future FMI's. Instructions are below.
 - Go to GIB: http://gib.gehealthcare.com/gib/gib_MainMenu.jsp
 - After the log in screen, click on [Global Installed Base entries]
 - Under "Serial Changes to Installed Base", enter the system ID of the system on which this coil is being installed. Click [Submit]
 - Click on [New Entries]
 - For both the Anterior and Posterior portions of the coil, enter the Model (Part number), Serial number, Installation Date and "A", then click [Submit].

SECTION 3: 32CH CARDIAC COIL MCQA TEST

Follow this procedure to perform the MCQA test. There are two phantom options that can be used for MCQA on the MR450 system. Only the unified phantom can be used on MR450w.

3.1 Tools Required

TOOLS REQUIRED - TABLE 3-1-1

Description	GE Part #	Qty
Legacy TL Phantom Brick (Not an option on MR450w) Or Unified TL Phantom Brick	2264744-8 Or 5343347	1
32Ch Cardiac Coil - Phantom Positioner (2 pieces)	5366627-4	1

3.2 MCQA Setup

The MCQA tool will be run separately for the posterior and the anterior coils. The coil positioning and the phantom setup for the MCQA is the same for both the coil components.

TABLE 3-2-1 Steps for Coil and phantom setup (Anterior or Posterior coil MCQA test)

Visual	Description
	1. Remove any other surface coils (if present) from the cradle.
 Illustration 6	2. Remove any pads present on the coil. The cable must point towards the magnet and plug into ports P1 or P2. Place lower half of phantom positioner on table. Ensure that the positioner is centered in the R/L direction. Insert coil half (anterior or posterior) and place top part of the positioner on the coil as shown in Illustration 6 and Illustration 7.



Illustration 7

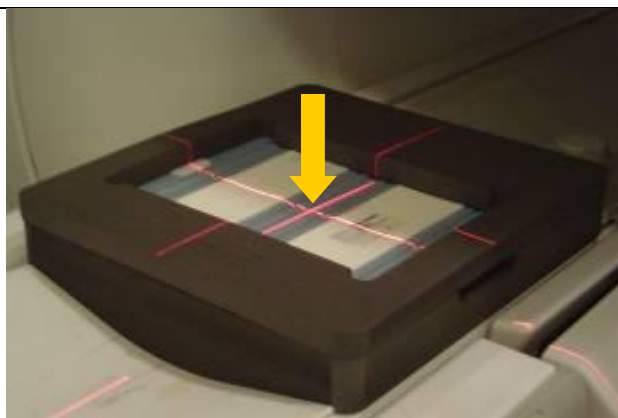


Illustration 8

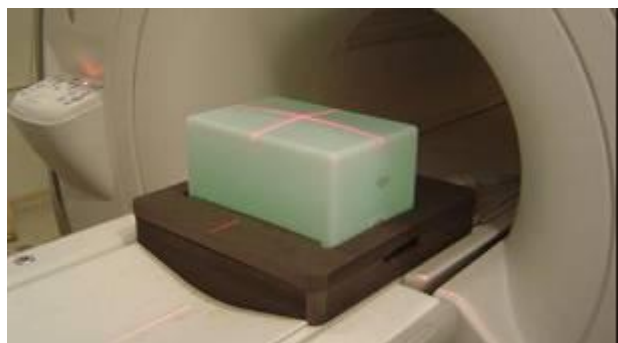


Illustration 9: MCQA SETUP WITH UNIFIED TL BRICK



3. Turn alignment button on

Move table into bore until landmark lasers intersect with crosshairs on the coil as shown in Illustration 8.

Press 'Landmark' button

Place phantom (legacy or unified TL brick) on the coil. The legacy phantom option is not available on MR450w systems. Ensure that the phantom is contained within the phantom positioner as shown in Illustration 9 and Illustration 10.

Press 'Advance to Scan' button.

Illustration 10: MCQA setup with legacy TL Brick

3.3 MCQA Test

3.3.1 At the operator workstation, end the previous exam (if required).

3.3.2 From the Common Service Desktop, select the [Image Quality] tab, choose “Multi-Coil QA Tool”, and then “Click here to start this tool” as shown in Illustration 11.

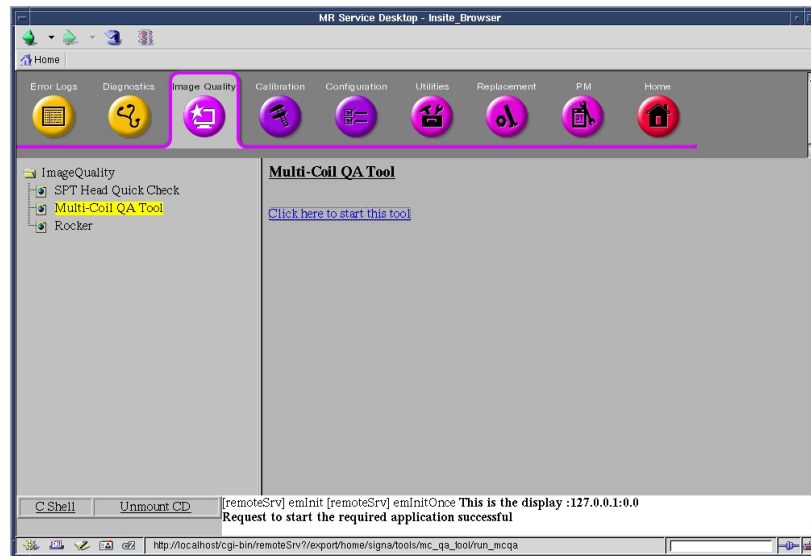
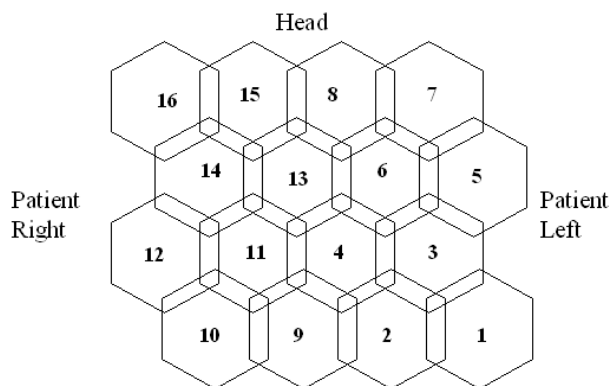


Illustration 11: MCQA from the Service Desktop

3.3.2 Further instructions on using the MCQA tool can be found on the MR450 or MR450w Service Methods CD or the web at: [Troubleshooting -> System -> Multi-Coil Quality Assurance Tool](#).

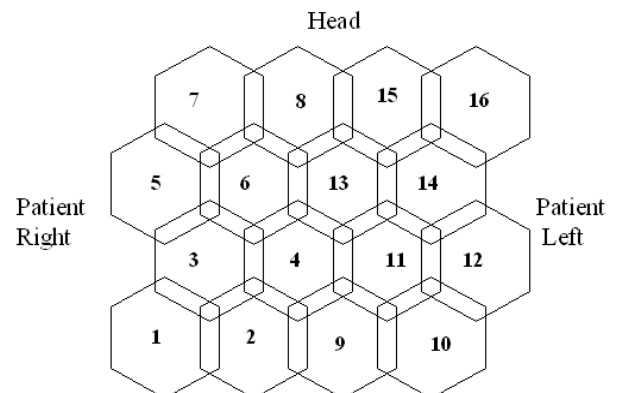
For troubleshooting: The relationship between MCQA image order, coil Receive Pin and coil element are the same (IMG 1 = Rx Pin 1 = Element 1). *Illustration 12* below shows the element layout for the anterior and posterior sections.

Posterior Section Viewed From Top (In positioning pad!)



Foot (cable side, also exits along patient right)
Hexagon diameter = 3.65" measured from center of trace

Anterior Section Viewed From Top



Foot (cable side also exits along patient right)

Illustration 12: Element layout for 32Ch Cardiac Coil

3.4 Periodic Quality Assurance Check

QA checks should be performed during installation and during planned maintenance. Perform the quality assurance checks as outlined below to ensure the coil is operating properly.

3.4.1 Check external cable for cracks or cuts.

3.4.2 Perform Section 3: 1.5T 32-CHANNEL CARDIAC COIL MCQA TEST

SECTION 4: TROUBLESHOOTING

Note: The Invivo 32-channel cardiac coil is not to be opened for any reason!

4-1 Mechanical Hardware Check

- Verify that there are no loops in the cables.
- Verify that there are no metal or ferromagnetic objects close to the coil, patient or magnet (i.e., safety pin, hair pin).
- Verify that the coil is properly positioned.

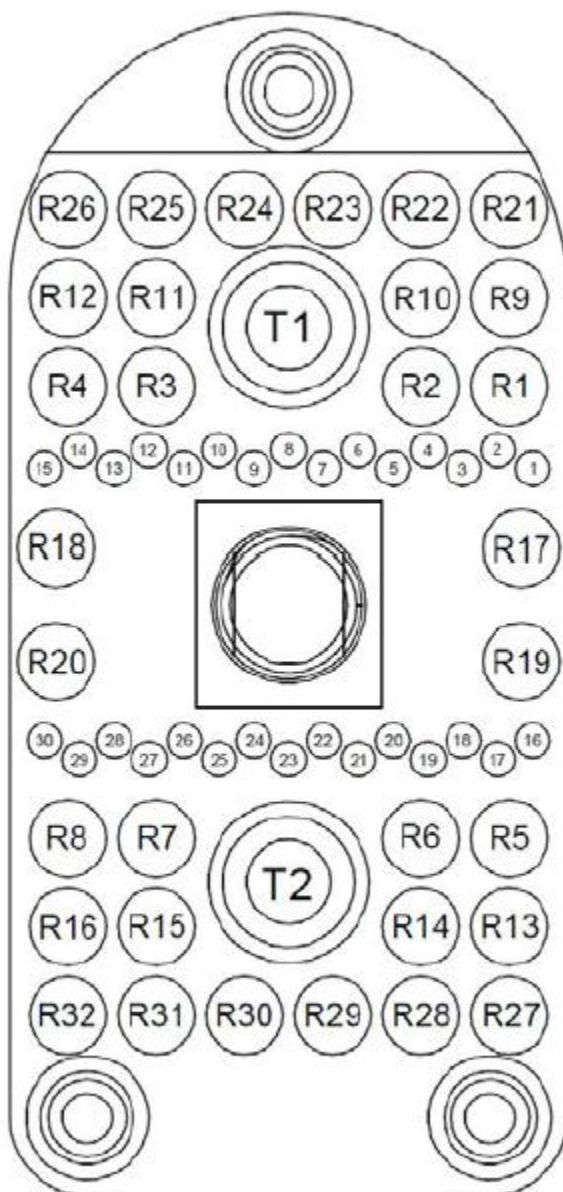
4-2 Electrical Hardware check

Visually inspect all coaxial connectors and pins on the system plug. If there are any broken, deformed or recessed pins or coaxial connectors, replace the coil component.

Coil Connector Illustration 13



Connector Pin Out Illustration 14



Connector pin out shown from the pin side of connector.

NOTE: The center pins of the System Connector *receive coaxial connectors* are extremely fragile [Connector Pin Out Illustration 14, R1-R8]. **Do not attempt to make any measurements at these center pins.**

For GND measurements, use any of the receive coaxial pins' outer shields (R1-R8).

RESISTANCE MEASUREMENTS – TABLE 4-2-1

System Connector		Reading
Positive Lead	Negative Lead	
1	2	< 8Ω
3	4	< 8Ω
5	GND	Open
6	GND	Open
20	GND	140 ±40Ω
21	GND	< 5Ω
22	GND	140 ±40Ω
23	GND	< 8Ω

If one or more of the readings are out of specification, replace the coil. If all of the above readings are correct, redo *Section 3: 32CH CARDIAC COIL MCQA TEST*. If the problem still exists, replace the 1.5T Cardiac Coil.

4-3 Coil ID Troubleshooting Tips

If the following error is displayed, there may be a problem with the Coil ID coil or system hardware: *The prescribed RF coil and detected RF coil do not match.*

Troubleshooting instructions are available through the Service Methods DVD or online via the path: [Troubleshooting](#) -> [Diagnostics](#) -> [Hardware Location](#) -> [Magnet Room](#) -> [Coil ID Functional Diagnostic](#)

The diagnostic will display the contents of the coil ID chip whenever the connector is inserted. Follow the troubleshooting instructions if the correct ID is not displayed. When the troubleshooting is complete, and if the correct ID is still not displayed, replace the coil.

SECTION 5 – MAINTENANCE

5-1 Coil Care

WARNING!

Detach coil connector from scanner before attempting to clean. Do not reattach after cleaning until coil has dried completely. Having the coil attached to the system during cleaning or when it is wet may result in electrical shock.

CAUTION

Do not spray or pour cleaning solution directly on coil. Do not submerge coil in solution. The coil contains sensitive electronics components that could be damaged by the solution.

The 1.5T 32-Channel Cardiac Coil, posterior pads must be cleaned and stored using the following procedure:

- Wipe with a cloth that has been dampened in a solution of 10% bleach and 90% tap water, or 30% isopropyl alcohol and 70% tap water.
- **Do not pour any cleaning solution directly on the coil!**
- Let the coil housing and pads dry before use.
- **Under no circumstances should the coil be placed into any type of sterilizer.**

SECTION 6 – RENEWAL PARTS

6.1 Field Replaceable Units

FIELD REPLACEABLE UNITS LIST – TABLE 6-1

Description	GE Part #	Invivo #
1.5T 32Ch Cardiac Coil - Anterior	5366664-2	4535-302-86221
1.5T 32Ch Cardiac Coil - Posterior	5366664-3	4535-302-86231
32Ch Cardiac Coil Phantom Positioner	5366627-4	4535-302-86261

6.2 Other Replaceable Accessories

The following table lists replaceable accessories that may be purchased from Invivo Corporation. The operator manual contains the contact information for Invivo.

OTHER REPLACEABLE ACCESSORIES LIST – TABLE 6-2

Description	Invivo #
32Ch Cardiac Coil Head Pad	4535-302-86281
32Ch Cardiac Coil Foot Pad	4535-302-86291
32Ch Cardiac Coil Anterior Pad	4535-302-86301
32Ch Cardiac Coil – Posterior Pad Includes: top posterior pad (Invivo part# 4535-302-58901), bottom posterior pad (Invivo part# 4535-302-58891), and straps (Invivo part# 4535-302-59791)	4535-302-86271
Straps Set (Includes both Anterior and Posterior Velcro straps)	4535-302-86311

Revision History

Rev	Date	Author	Primary Reason for Change	DCN Number
A	10-02-09	J. Ciesla	Initial Release	
B	10-12-09	Hohmuth	Updates	
C	10-28-09	Hohmuth	Update pics, index table	
D	12-3-09	J. Ciesla	Added install and MCQA sections	
E	12-10-09	J. Ciesla	Added element layout, edited required conditions for install	
F	02/05/10	J. Ciesla	Corrected part numbers	6809
01	02/05/10	J. Ciesla	Issued	6809
02	02/05/10	J. Ciesla	Corrected FRU numbers & label pictures	6809
03	04/01/10	J.Ciesla	Corrected Resistance Measurement in Table 4-2-1	7520